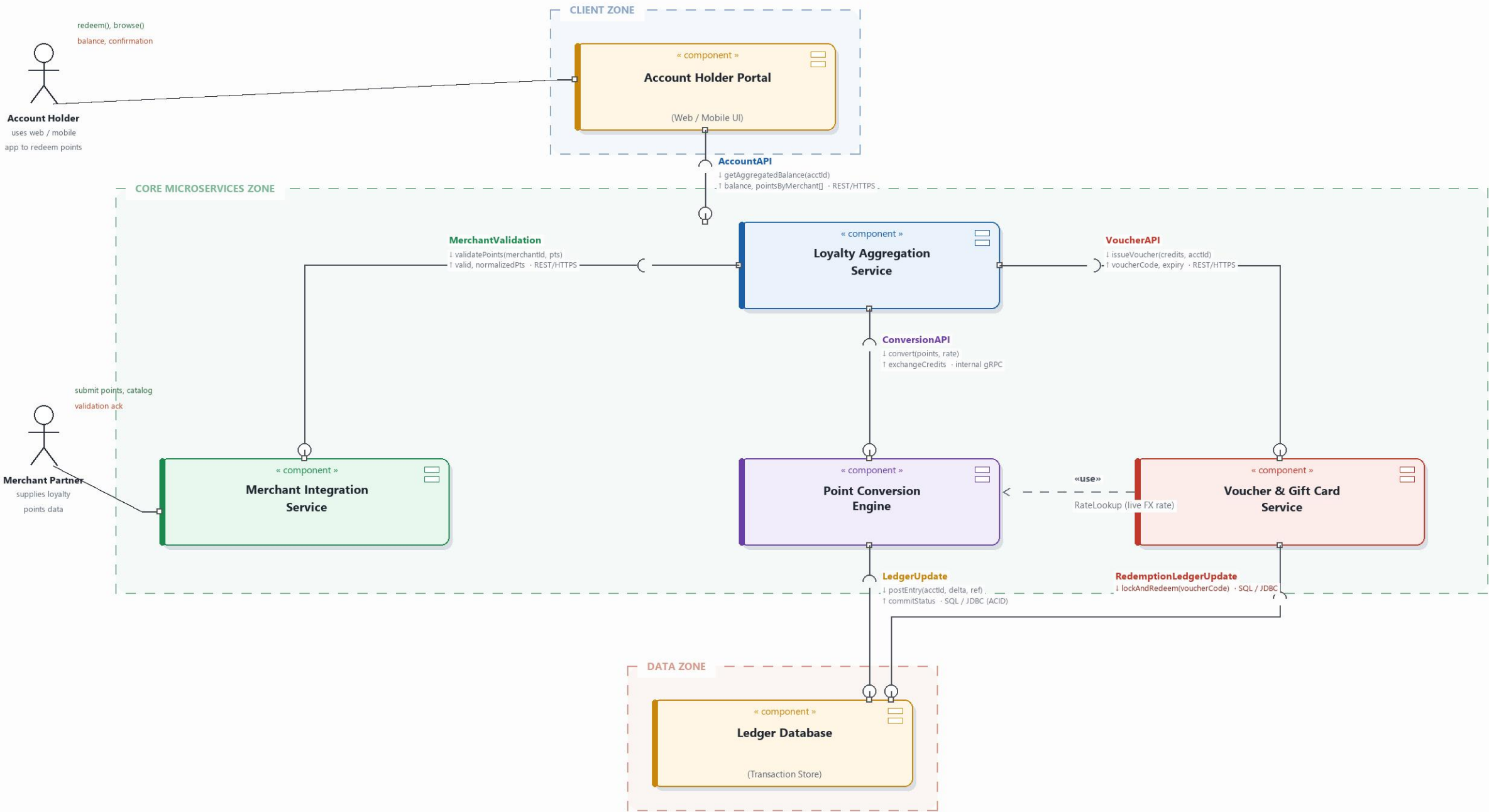


Digital Gift Card & Loyalty Point Exchange

Component Diagram — Microservices Architecture

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Problem Statement #34 • Actors: Account Holder, Merchant Partner



Legend

- Provided Interface (ball)
- Port (boundary point)
- ⌋ Required Interface (socket)
- > «use» dependency (dashed, points to supplier)
- «component» — modular, replaceable, black-box unit
- Actor — external stakeholder (Account Holder / Merchant Partner)

Architecture Notes

- Style: Microservices — each service owns one responsibility and its own data access path.
- Loyalty Aggregation Service is the orchestration entry point; it is the only service the Account Holder Portal talks to directly — downstream services stay hidden from clients.
- Voucher & Gift Card Service is isolated behind its own interface for anti-fraud auditing; a breach elsewhere cannot directly reach voucher issuance or the Ledger Database.
- Point Conversion Engine and Voucher Service scale independently to hold the 250 ms NFR-001 target under peak redemption load, without scaling lighter services.
- Any component can be replaced by another implementing the same provided interface.